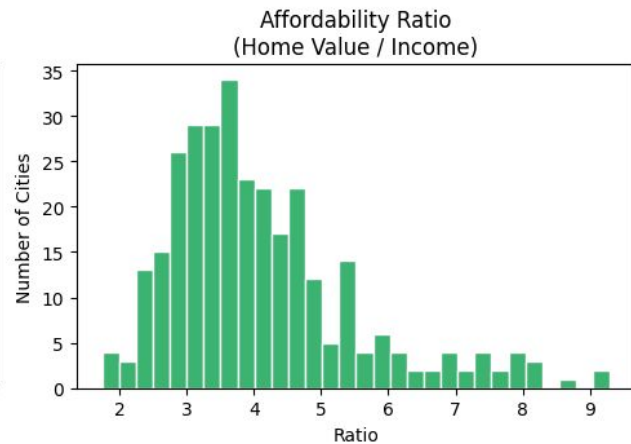
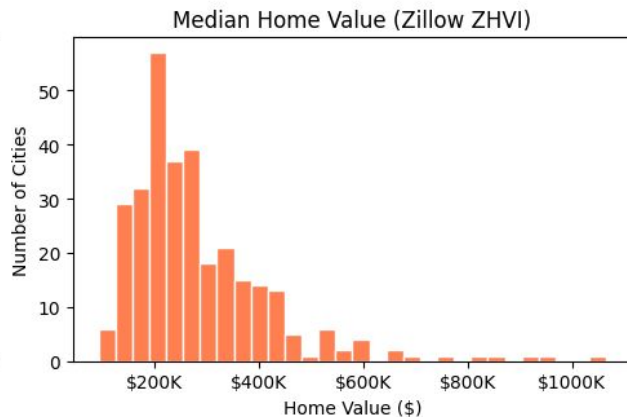
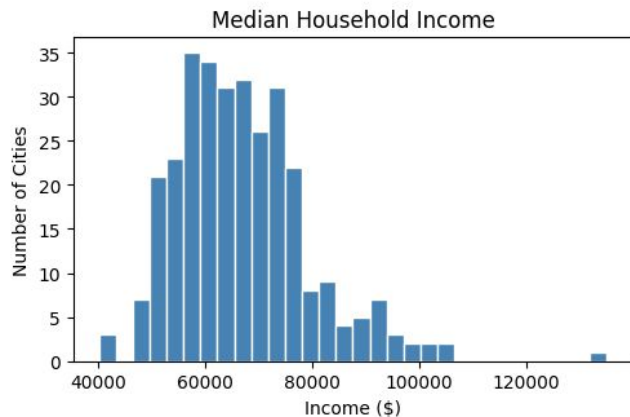


Can higher incomes keep up with rising home prices across U.S. cities?

Riley Prell

Median Household Income, Median Value of Homes, and Cities Affordability Ratio

Distribution of Key Housing Variables

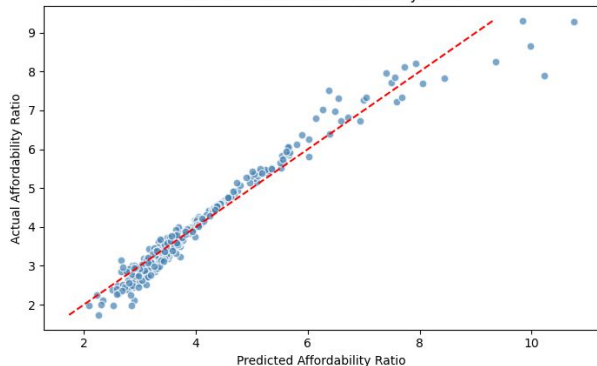


Can higher incomes keep up with rising home prices across U.S. cities?

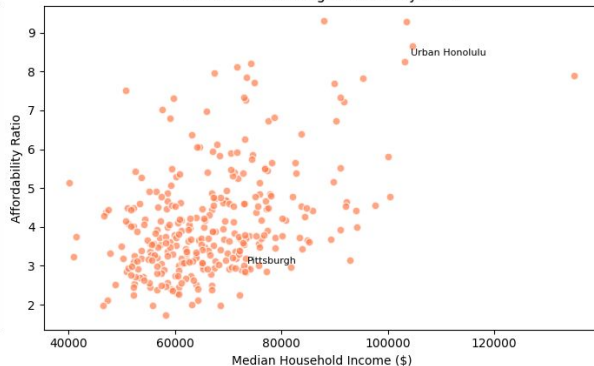
$$\text{AffordabilityRatio} = \beta_0 + \beta_1(\text{MedianIncome}) + \beta_2(\text{HomeValue}) + \varepsilon$$

Housing Affordability Model

Predicted vs. Actual Affordability Ratio



Income vs. Housing Affordability Ratio



Higher home prices make cities less affordable. While higher incomes help in cities like Honolulu and San Jose, home prices have risen so far that even high earners are priced out

95.1% variation in affordability explained

−0.60 drop in affordability ratio per \$10K income increase

+1.30 rise in affordability ratio per \$100K home price increase

The model struggles with outlier cities, these tend to be places on the coast where the prices skyrocket.

Residual Plot — OLS Assumptions Check

